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Graph Analytics

Basic Theory and Applications

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Outline

- Overview on Graphs
- Path Analytics
- Connectivity Analytics
- Community Analytics
- Centrality Analytics
- Pattern Matching
- Parallel Programming Model for Graphs
- Applied Graph Analytics
- Useful Resources

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Introduction

Graph Analytics and Databases

Graph Analytics - *"Built on the mathematics of graph theory, graph analytics help to understand, codify, and visualize* ***relationships that exist between objects*** *in a given domain context, in order to* ***uncover insights*** *about the* ***structures and patterns*** *of the objects relationships."*

Graph Databases – *"A NoSQL family of data stores that is* ***optimized to store, model, and process data in a graphical form****, as well as answering graph-related* ***queries efficiently****."*

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Graphs Overview

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What is NOT a Graph?

Basic Concepts

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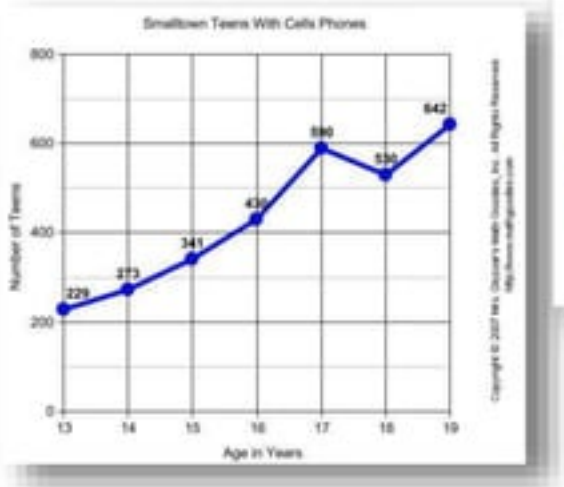
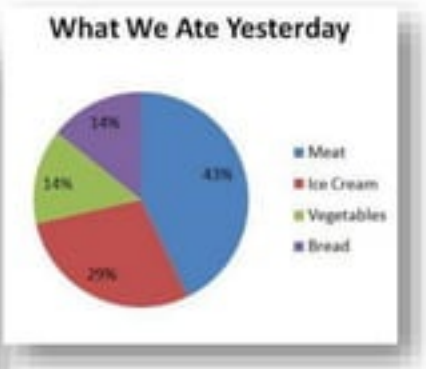
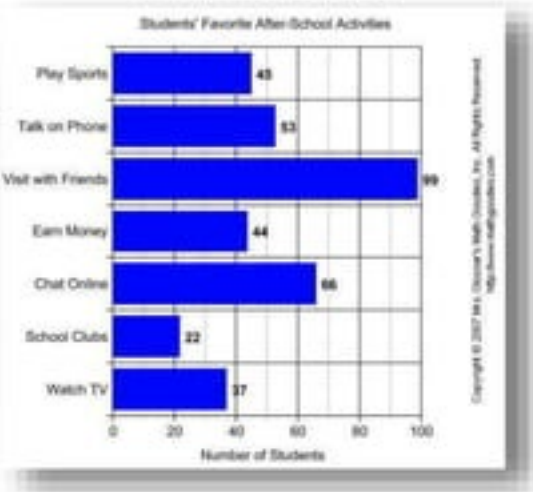
These are **NOT** graphs!







These are **charts!**

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What is a Graph?

Basic Concepts

*In computing, a graph is abstract **data structure** that represents set objects and their relationships as **vertices and edges**, and supports a number of graph-related **operations***

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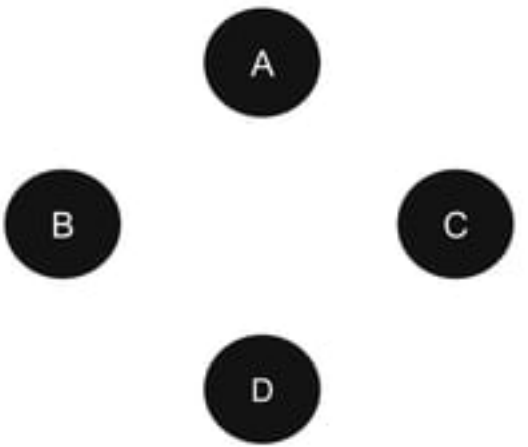
- **Objects (nodes)**: {A, B, C, D}



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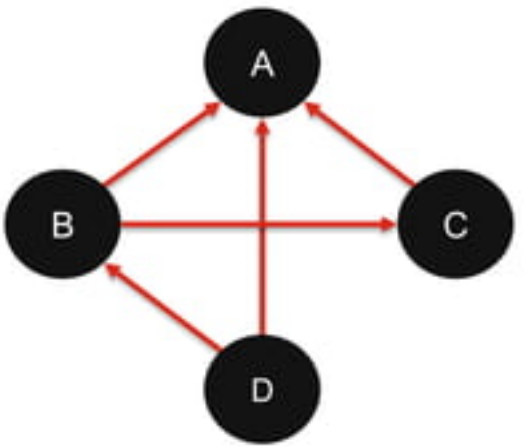
- **Objects (nodes)**: {A, B, C, D}
- **Relationships (edges):** {(D,B),(D,A),(B,C),(B,A),(C,A)}



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What is a Graph?

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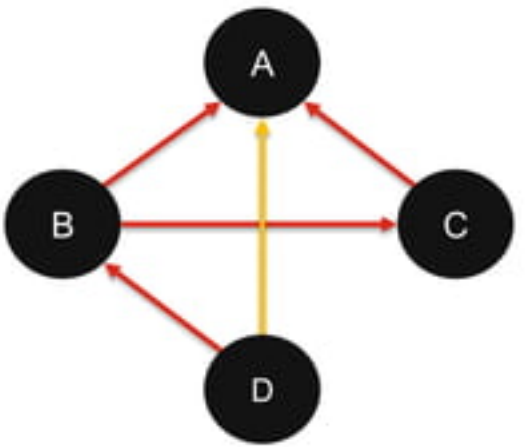
- **Objects (nodes)**: {A, B, C, D}
- **Relationships (edges):** {(D,B),(D,A),(B,C),(B,A),(C,A)}
- **Operation**: shortest path between D and A



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What is a Graph?

Graph operation examples

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- graph.GetNodes(<condition>)
- graph.GetEdges(<condition>)
- graph.AddNode(node)
- graph.AddEdge(node1,node2)
- graph.AddEdge(edge)
- graph.RemoveNode(node)
- graph.GetShortestPath(node1,node2)
- graph.Neighbours(node,level)
- graph.GetDistance(node1,node2)

- node.GetParents()
- node.GetChildren()
- node.GetAncestors(level)
- node.GetDescendants(level)
- node.IsAncestorTo(node2)
- node.IsDescendant(node2)
- node.AddParent(parentNode)
- node.AddChild(childeNode)
- node.IsReachable(node2)

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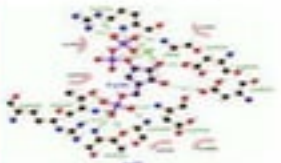
What is a Graph?

Real-world examples…

Social Media – Twitter

- Identify groups (communities) and group interactions
- Find influencers in community
- Extract topic interests

Biology – Biological Entities Networks



- Discover unknown relationships (gene/ protein to disease, disease to disease, cure to disease, etc.)
- Exploratory Data Analysis & anomaly detection

Geo IS – Smart Cities

- Coverage analysis
- Traffic flow, congestion estimation, routing
- Failure Impact analysis

Reasoning – Predictive Maintenance

- Predict the next state given the current (and previous state(s))
- Compute the probability of sequence of event

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Why Graphs?

Importance of graph data structures

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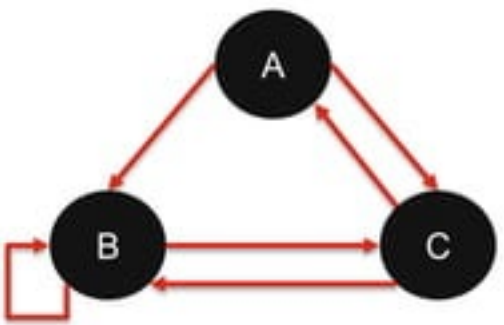
Graph Types

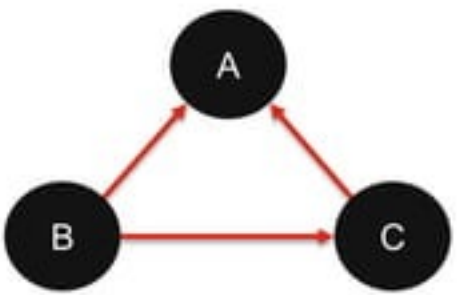
Directionality and circulation

Directed Graphs



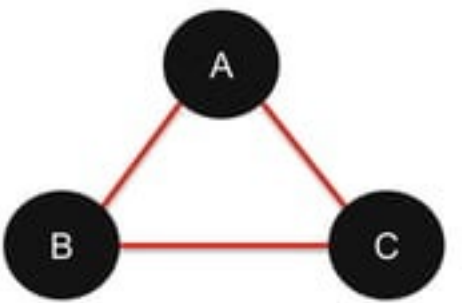
State-transition models

Directed Acyclic Graphs



Dependency networks

Undirected Graphs



Connectivity networks

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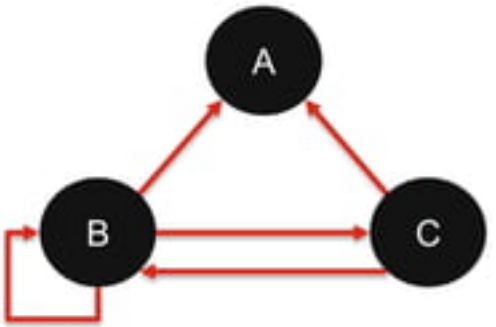
Simple Graph Representation

Adjacency Matrix



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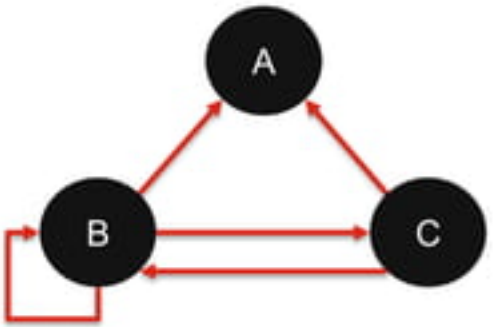
Simple Graph Representation

Adjacency Matrix



To

From	A	B	C
A	0	0	0
B	1	1	1
C	1	1	0

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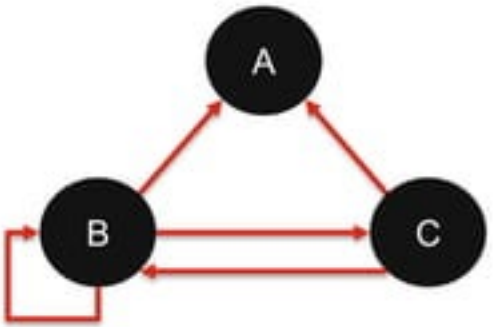
Simple Graph Representation

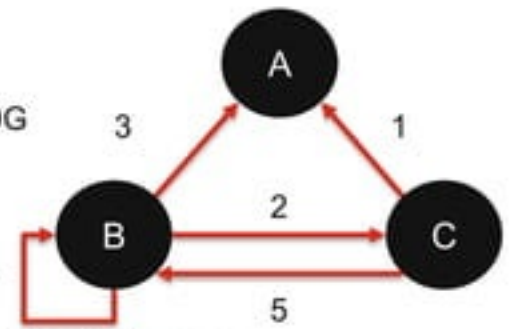
Adjacency Matrix



To

From	A	B	C
A	0	0	0
B	1	1	1
C	1	1	0

Weighted DG



From	A	B	C
A	0	0	0
B	3	4	2
C	1	5	0

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Simple Graph Representation

Edge Table

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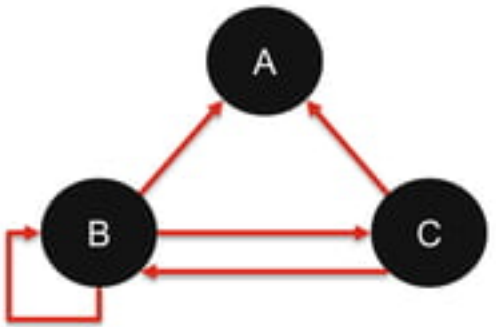

FROM	TO	WEIGHT
B	A	..
B	C	..
B	B	..
C	B	..
C	A	..

Useful in Relational Databases

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Simple Graph Representation

Adjacency list

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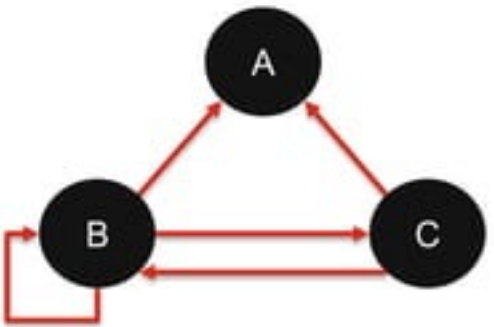

Node	IN	OUT
A	B,C	-
B	B,C	A,C
C	B	A,B

Useful in MapReduce

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Label Property Graph Model

Defining information-rich graphs

In a simple model, a graph consist of:

- A set of vertices (nodes)
- A set of edges (each connecting two nodes)

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In the Label Property Graph Model, each element (vertex/edge) has:

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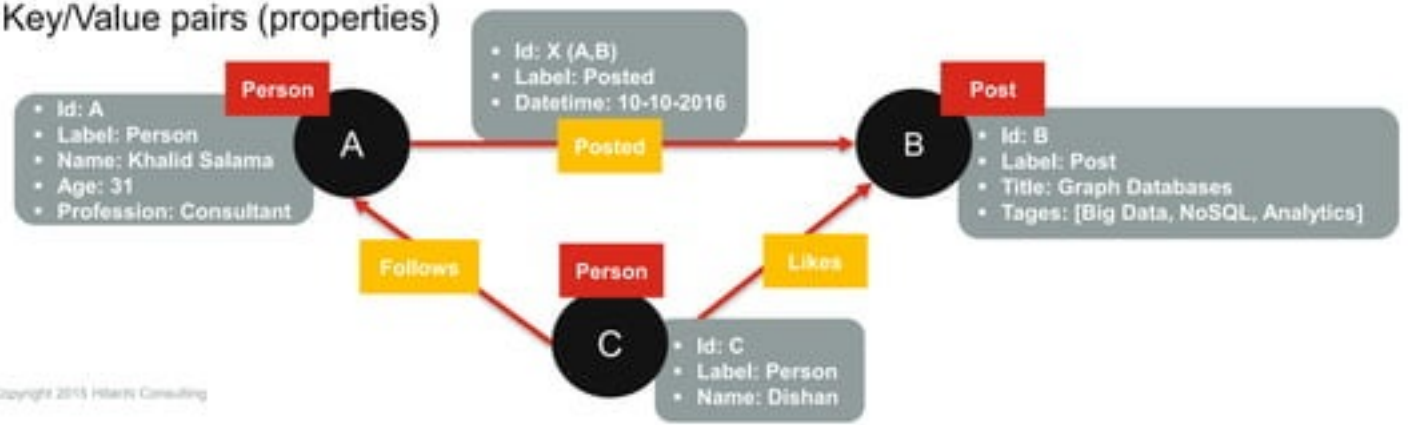

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Types of Graphs Analytics

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Types of Graph Analytics

Relationships Analytics

Connectivity Analytical

Path Analytics & Traversing

Community Analytics

Centrality Analytics

Pattern Matching

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Connectivity Analytics

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Connectivity Analytics

Graph structural analysis

How big is the graph?

Volume – Number of edges increases quadraticly with respect to number of nodes

Velocity – How frequent a new vertex or edge is added to the graph

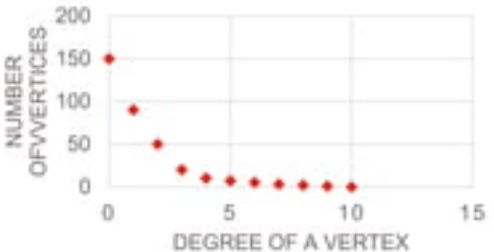
Degree

- In-degree of a vertex: number of edges pointing to the vertex (parents)
- Out-degree of a vertex: number of edges point out of the vertex (children)
- Degree of a vertex: number of neighbour of a vertex in an undirected graph

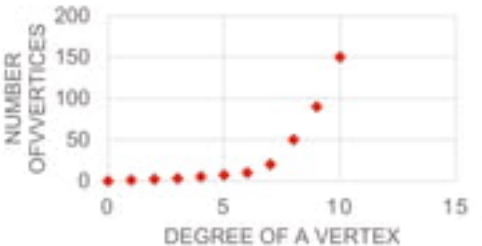
Connectivity Analytics

Graph structural analysis

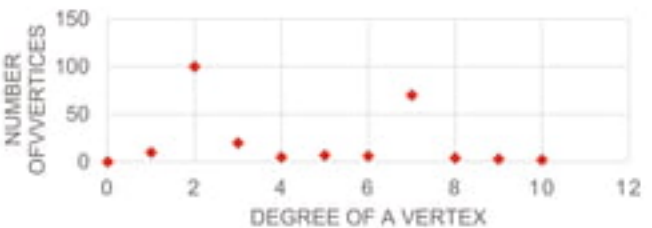
Degree Histogram – describes the skewness of the degree distribution in a graph



Exponentially unlikely to find a vertex with increased degree



In some case, it is more likely to find more vertices with high number of edges



Or it can be multi-modal

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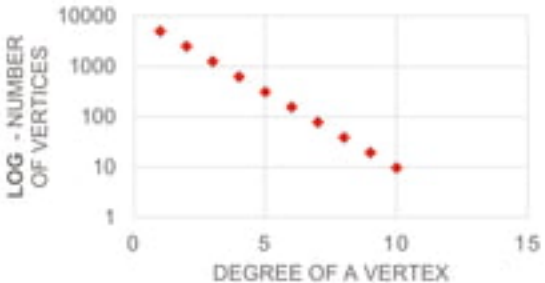
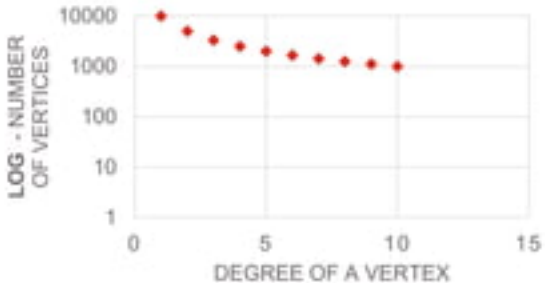
Graph structural analysis

Degree Histogram – Random vs Natural Graphs



In random graphs, exponentially unlikely to find a vertex with increased degree

Exponential Distribution $n(d) \simeq c\left(\frac{1}{2}\right)^d$



In some case, it is more likely to find more vertices with high number of edges

ZipF Distribution $n(d) \simeq \frac{1}{d^r}$

A vertex with higher degree (more connections) is more likely to get a new edge, compared to less connected vertices – Social Networks

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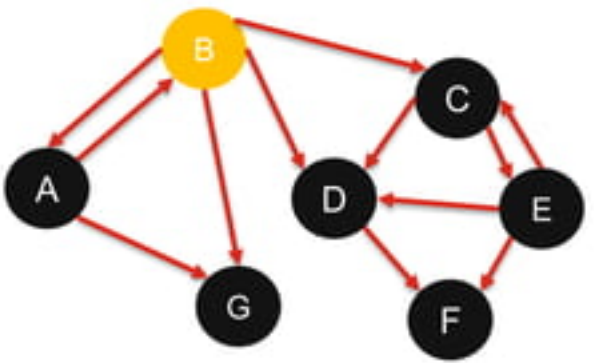
Connectivity Analytics

Graph structural analysis

- **Highly connected nodes** – nodes with high In/Out-Degree.
- **Graph Robustness** – how easy to break the graph by removing a few nodes/edges (Built-in Redundancy)
  - **Connectivity Coefficient**: minimum number of nodes you need to remove to disconnect a graph (E.g. node B)
    - Useful in network fragility analysis and social media advertising
  - **Connectivity**: X is reachable from Y **OR** Y is reachable from X
  - **Strong Connectivity**: X is reachable from Y **AND** Y is reachable from X
    - High degree nodes make the network more vulnerable.

- Graph Comparison – how similar graph G1 to G2?
  - Number of nodes
  - Number of edges
  - Ratio of Nodes to Edges
  - In/Out Degree Histogram
  - Connectivity Coefficient



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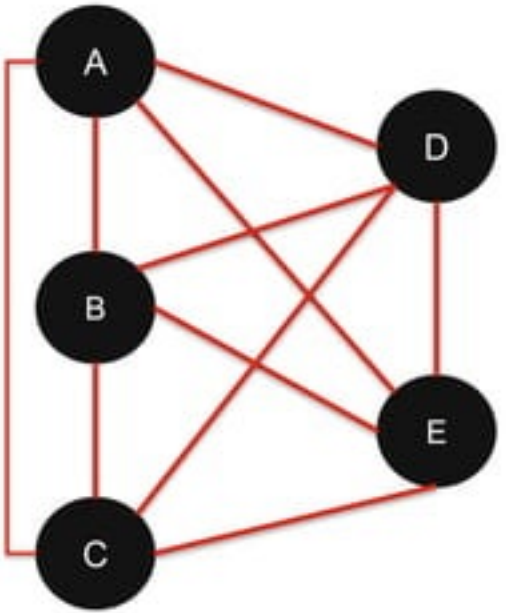
Connectivity Analytics

Graph structural analysis

- **Fully connected graph**: Each node has edges to all the other nodes (usually undirected graph)
  - Can we find subgraphs, in a given graphs, that are fully connected? **(Cliques)**



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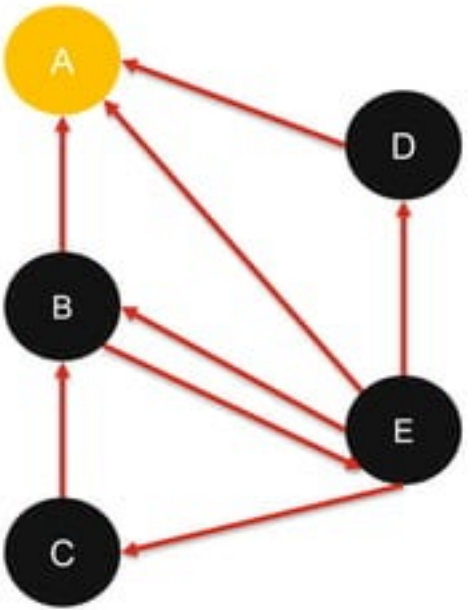
Connectivity Analytics

Graph structural analysis

- **Fully connected graph**: Each node has edges to all the other nodes
- **Terminal node**: A node with no outgoing edges



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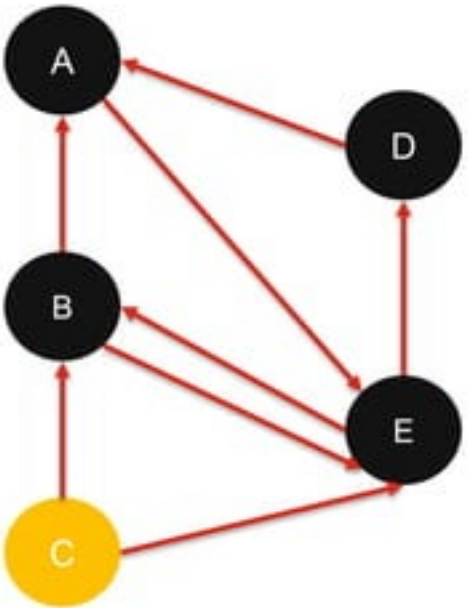
Connectivity Analytics

Graph structural analysis

- **Fully connected graph**: Each node has edges to all the other nodes
- **Terminal node**: A node with no outgoing edges
- **Unreachable node**: A node no ingoing edges



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Connectivity Analytics

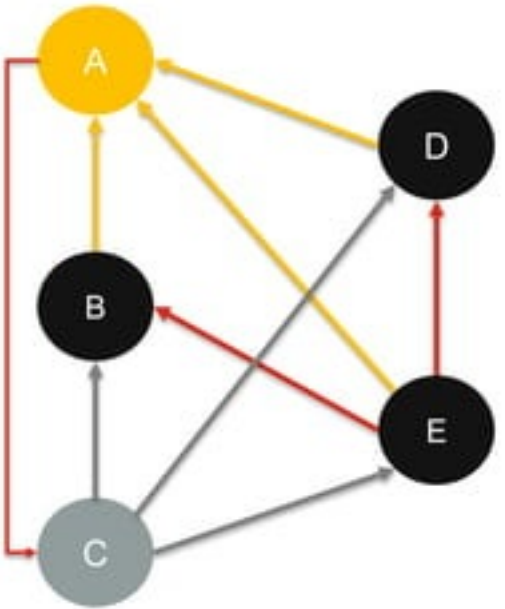
Graph structural analysis

- **Fully connected graph**: Each node has edges to all the other nodes
- **Terminal node**: A node with no outgoing edges
- **Unreachable node**: A node no ingoing edges
- **Hub vs. Authorities:** High In-degree vs High Out-degree

  A is a hub node, C is an authority node

  E.g.: Social Networks: Talkers vs. Listener

  E.g.: Web structure



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Path Analytics

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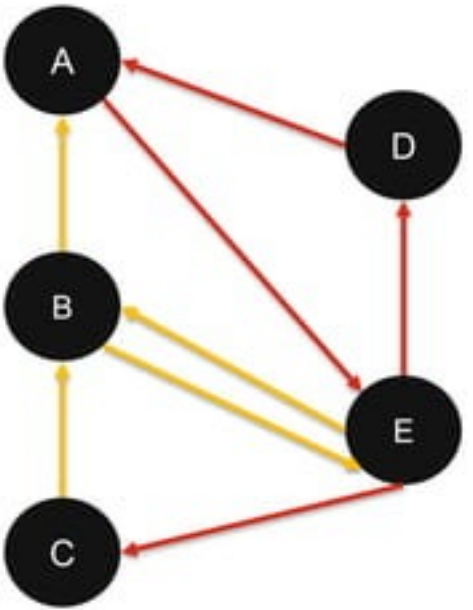
Path Analytics & Graph Traversing

Concepts and operations

- **Path**: A set of (ordered) edges between node x and node y

  C → B → E→ B → A



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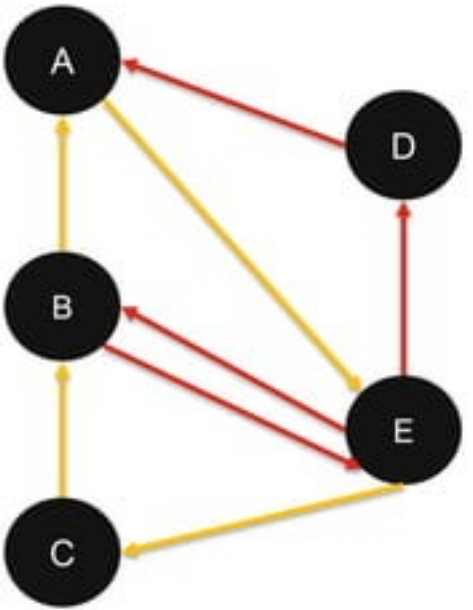
Path Analytics & Graph Traversing

Concepts and operations

- **Path**: A set of (ordered) edges between node x and node y
- **Cycle**: A path where the start and the end nodes are the same

  E→C→B→A→E



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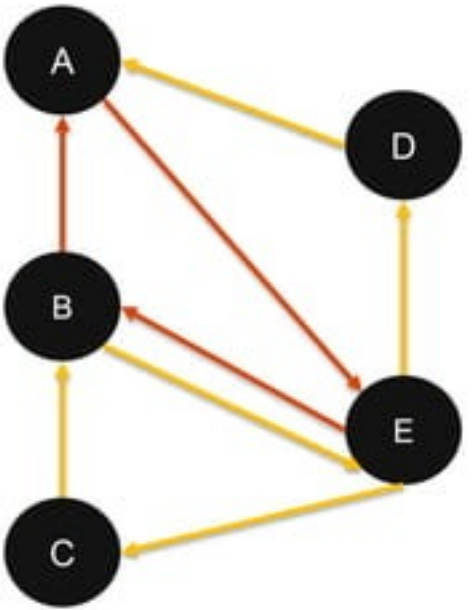
Path Analytics & Graph Traversing

Concepts and operations

- **Path**: A set of (ordered) edges between node x and node y
- **Cycle**: A path where the start and the end nodes are the same
- **Trail**: A path with no repeated edges

  E→C→B→E→D→A



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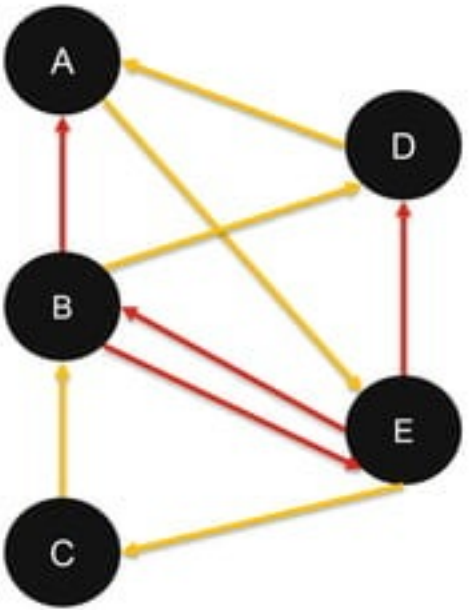
Path Analytics & Graph Traversing

Concepts and operations

- **Path**: A set of (ordered) edges between node x and node y
- **Cycle**: A path where the start and the end nodes are the same
- **Trail**: A path with no repeated edges
- **Tour**: A cycle traversing all the nodes, only once.

  D→A→E→C→B→D



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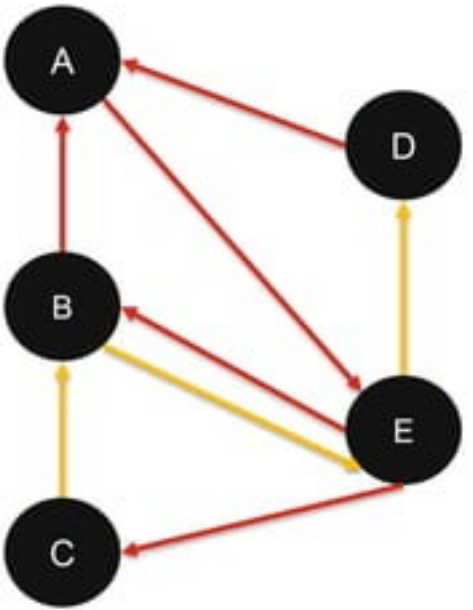

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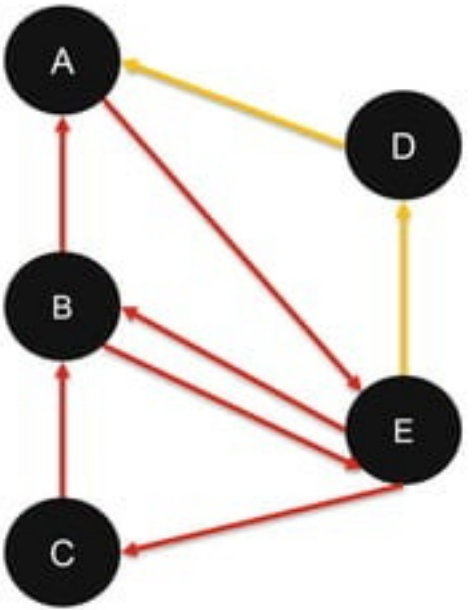

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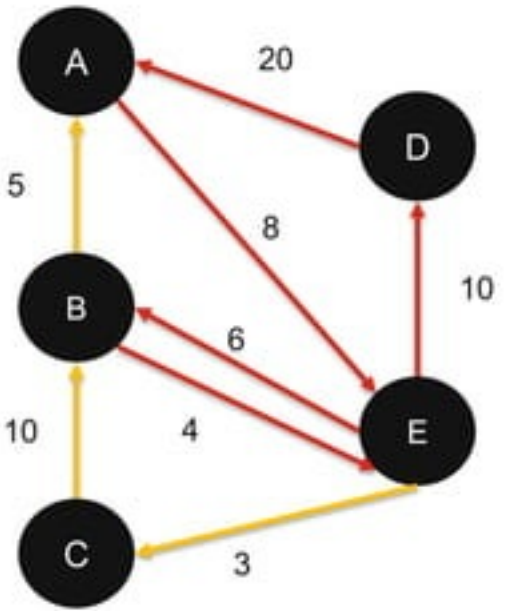
  - Optimize a given function
  - Satisfy given constrains



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- **Graph Diameter**: The longest "shortest path" between two (reachable) nodes (Distance Matrix) – Structural Analysis

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Distance Matrix
(Shortest Path Paris)

	A	B	C	D	E
A	-	8	∞	10	5
B	8	-	12	13	∞
C	11		-	20	∞
D	4	9	16	-	10
E	7	∞	∞	5	-

In this example (Directed Graph), the Graph Diameter is 20, which is the longest shortest path (that is the one from C to D)

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Path Analytics & Graph Traversing

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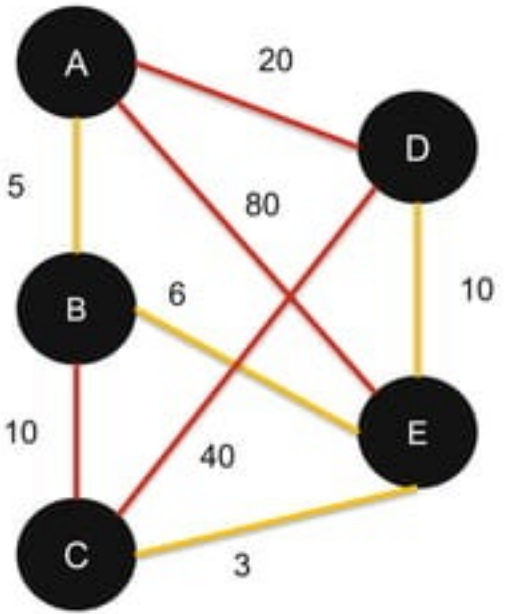
- **Minimum Spanning Trees**: edges that connect all the nodes with no cycles and minimum weight.



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Community Analytics

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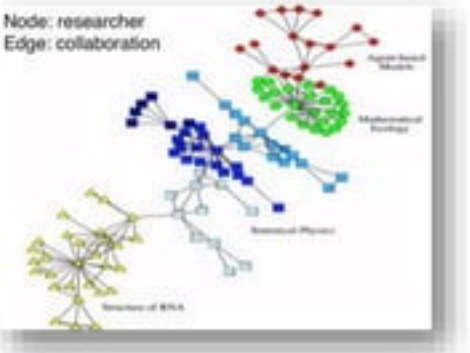
Community Analytics

Graph Clustering/Partitioning

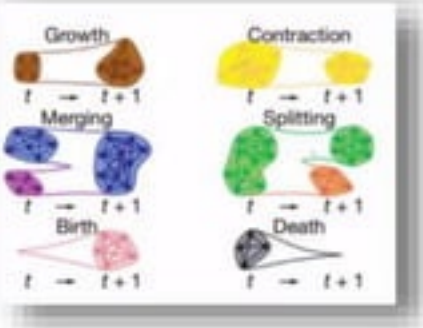
A dense subgraph (cluster) with in a graph, in which its nodes are more connected with a cluster than to the nodes outside the cluster

- Cohesion – Connectivity "within" the cluster is high
- Separation – Connectivity "between" clusters is low



Analytical Questions

- **Static** – Discover community
- **Static** – Describe interaction with a community
- **Static** – Describe interaction between communities
- **Temporal** – How a community emerged/dissolved?
- **Temporal** – Which communities are stable
- **Temporal** – Predict of a node will migrate to another community?



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Community Analytics

Graph Clustering/Partitioning

Finding Communities

Local Properties

- **n-Clique** (distance): largest subgraph that the maximum distance between each two nodes is <= n
- **n-Clans** (distance): an n-clique in which the largest distance between nodes in the subgraph is <=n
- **k-Core** (density): largest subgraph that each nodes is connected to at least k-nodes within the sub graph

Global Properties

Modularity –

- The fraction of the edges that fall within the given subgraph minus the expected such fraction if edges were distributed at random
- Reflects the **concentration of edges** within subgraph **compared with random distribution of edges** between all nodes regardless of subgraphs.

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Centrality Analytics

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Centrality Analytics

Vertex Importance Analysis

Connectivity Importance

- Simply, the **degree of node** X (in and **out** degrees).
  - I.e., the queen bees in a community (used for target marketing)

Closeness Importance

- Average **length** of all its **shortest paths**, compared to the averages of the other vertices (using Distance Matrix)
  - I.e., From vertex X, you can reach most of the other vertices quicker

Betweeness Importance

- The **fraction** of the **shortest paths** that X appears in.
  - I.e., if x is important, then most of the (shortest) paths between any two vertices in a graph pass through x (important underground station).

Vulnerability

- Vertex X belongs to the minimum node set that, if removed from the graph, the graph is **disconnected**.
  - Or, its removal will cause a high disruption in the network

Network Centralization (graph-level measure) – Measure of degree of variation of centrality score amongst the nodes of the network

Centrality Analytics

Vertex Importance Analysis

Page Rank - The importance (rank) of a vertex is computed as the total rank of all its adjacent edges (a.k.a Eigenvector Centrality).

- I.e., the importance of a given vertex is **not only how well-connected** it is, it is also **how well-connected its neighbours** are.
- Including a damping factor: the further the you go away the vertex, the less important it is on the rank of the vertex

PageRank can be interpreted as the probability to visit a page…

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Pattern Matching

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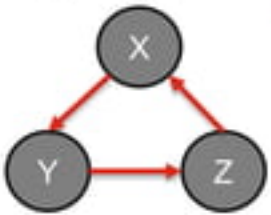
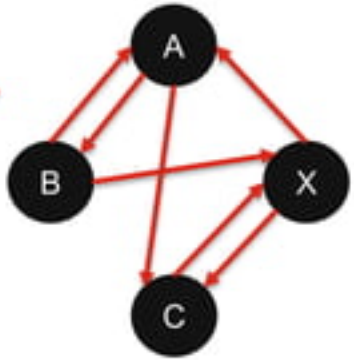
Pattern Matching

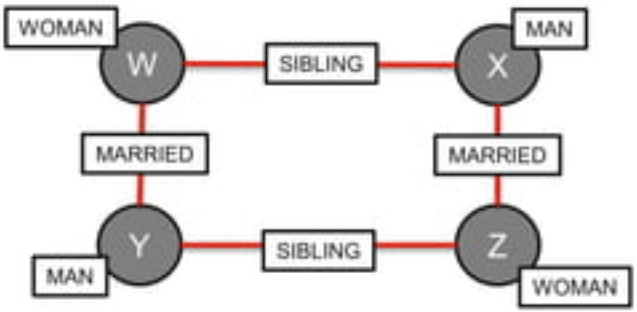
Graph Query

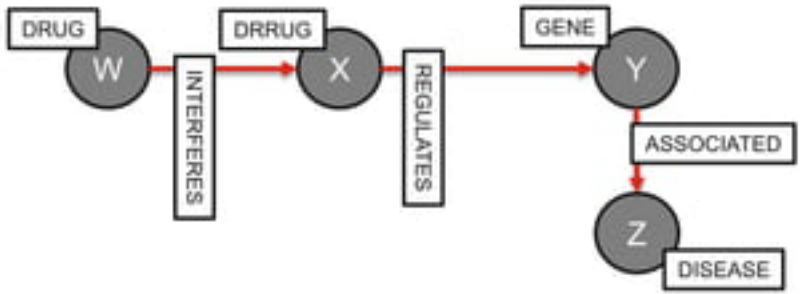
- Find the following patterns in a given graph





- Find the following patter in a given Property Model graph





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Pattern Matching

Applications

- **Banking** – Fraud Detection
- **Security** – Threat Detection
- **Bioinformatics & Biochemistry** – Association Analysis
- **Social Networks** – Job/Candidate suggestion
- **GPS & Smart Cities** – Traffic/Accident Analysis
- **Telecom** – Targeted Campaigning

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Parallel Programming Model for Graphs

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Parallel Programming Model for Graphs

Graph Processing

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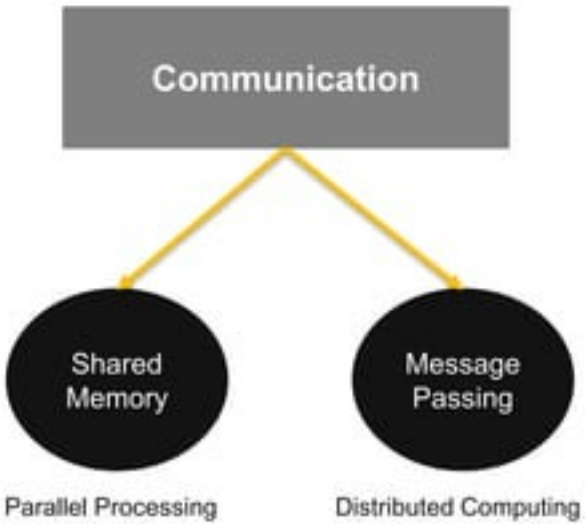

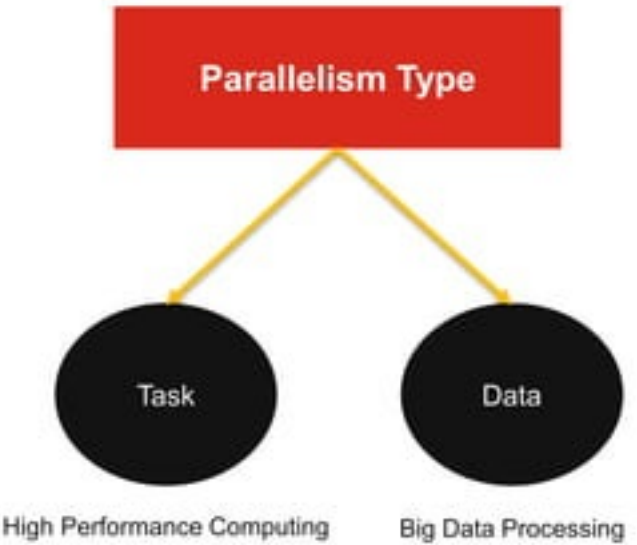

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Parallel Programming Model for Graphs

Graph Processing

- **Data Parallelism –** Each compute node has a subset of graph vertex.
- **Message Passing –** A vertex can communicate (send/receive a message) to a vertex (in another compute node) if it has an outgoing edge to.
- **Processing of vertices is performed in parallel** – E.g., Bulk Synchronous Parallelism (BSP)

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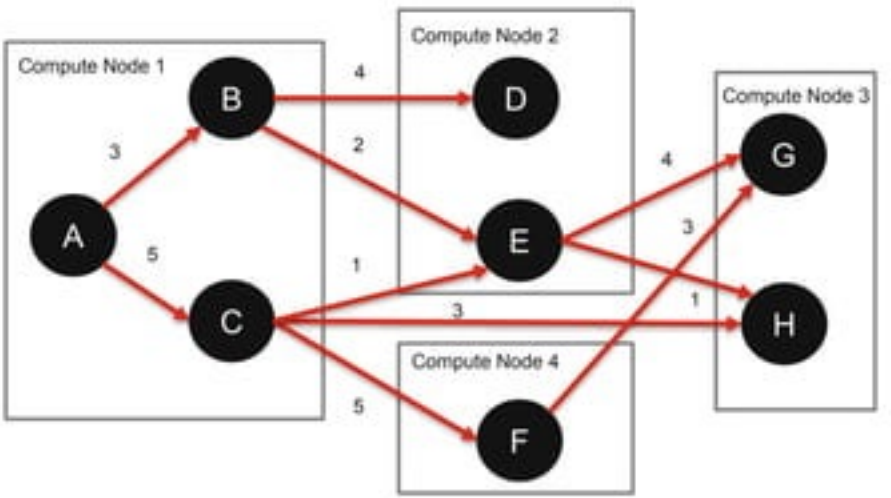
E.g.: Find the shortest path between A, H, in parallel



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Parallel Programming Model for Graphs

Graph Processing

Pregel - A System for Large Scale Graph Processing

- Published by Google
- Based on Bulk Synchronous Parallelism
  - Receive Messages from parent Nodes
  - Compute
  - Send Messages to Child Nodes
  - Pause & Synchronize

  } Superstep
- Example Application: PageRank

Graph Processing Tools:

- **Giraph –** HDFS, MapReduce, YARN (JAVA)
- **GraphX** – Spark, RDDs (Scala)

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Parallel Programming Model for Graphs

Graph Processing - GraphX

```
class Graph[VD, ED] {
  // Information about the Graph ===================================
  val numEdges: Long
  val numVertices: Long
  val inDegrees: VertexRDD[Int]
  val outDegrees: VertexRDD[Int]
  val degrees: VertexRDD[Int]
  // views of the graph as collections =============================
  val vertices: VertexRDD[VD]
  val edges: EdgeRDD[ED]
  val triplets: RDD[EdgeTriplet[VD, ED]]
  // functions for caching graphs ==================================
  def persist(newLevel: StorageLevel = StorageLevel.MEMORY_ONLY): Graph[VD, ED]
  def cache(): Graph[VD, ED]
  def unpersistVertices(blocking: Boolean = true): Graph[VD, ED]
  // change the partitioning heuristic ============================
  def partitionBy(partitionStrategy: PartitionStrategy): Graph[VD, ED]
  // transform vertex and edge attributes ==========================
  def mapVertices[VD2](map: (VertexId, VD) => VD2): Graph[VD2, ED]
  def mapEdges[ED2](map: Edge[ED] => ED2): Graph[VD, ED2]
  def mapEdges[ED2](map: (PartitionID, Iterator[Edge[ED]]) => Iterator[ED2]): Graph[VD, ED2]
  def mapTriplets[ED2](map: EdgeTriplet[VD, ED] => ED2): Graph[VD, ED2]
  def mapTriplets[ED2](map: (PartitionID, Iterator[EdgeTriplet[VD, ED]]) => Iterator[ED2])
    : Graph[VD, ED2]
  def reverse: Graph[VD, ED]
  def subgraph(epred: EdgeTriplet[VD,ED] => Boolean = (x => true),
      vpred: (VertexId, VD) => Boolean = ((v, d) => true)): Graph[VD, ED]
```

```
  // Modify the graph structure ====================================
  def mask[VD2, ED2](other: Graph[VD2, ED2]): Graph[VD, ED]
  def groupEdges(merge: (ED, ED) => ED): Graph[VD, ED]
  // Join RDDs with the graph ======================================
  def joinVertices[U](table: RDD[(VertexId, U)])(mapFunc: (VertexId, VD, U) => VD)
        : Graph[VD, ED]
  def outerJoinVertices[U, VD2](other: RDD[(VertexId, U)])
      (mapFunc: (VertexId, VD, Option[U]) => VD2) : Graph[VD2, ED]

// Aggregate information about adjacent triplets ===================
  def collectNeighborIds(edgeDirection: EdgeDirection): VertexRDD[Array[VertexId]]
  def collectNeighbors(edgeDirection: EdgeDirection): VertexRDD[Array[(VertexId, VD)]]
  def aggregateMessages[Msg: ClassTag](
      sendMsg: EdgeContext[VD, ED, Msg] => Unit,
      mergeMsg: (Msg, Msg) => Msg,
      tripletFields: TripletFields = TripletFields.All)
    : VertexRDD[A]
  // Iterative graph-parallel computation ==========================
  def pregel[A](initialMsg: A, maxIterations: Int, activeDirection: EdgeDirection){
      vprog: (VertexId, VD, A) => VD,
      sendMsg: EdgeTriplet[VD, ED] => Iterator[(VertexId,A)],
      mergeMsg: (A, A) => A)
    : Graph[VD, ED]
  // Basic graph algorithms ========================================
  def pageRank(tol: Double, resetProb: Double = 0.15): Graph[Double, Double]
  def connectedComponents(): Graph[VertexId, ED]
  def triangleCount(): Graph[Int, ED]
  def stronglyConnectedComponents(numIter: Int): Graph[VertexId, ED]
}
```


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Applied Graph Analytics

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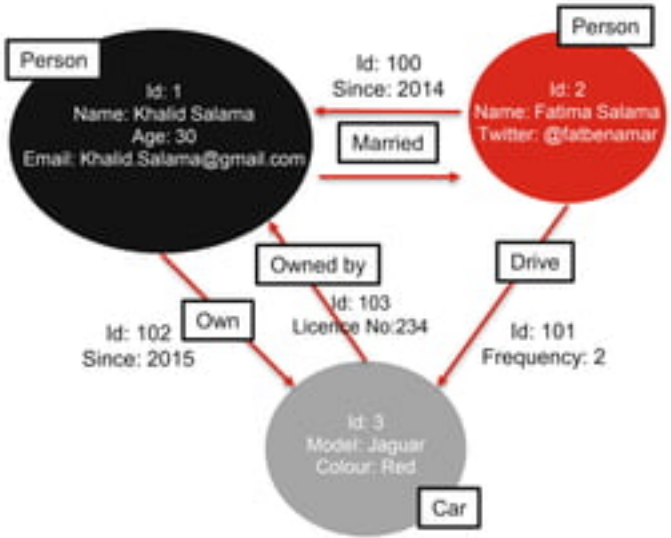
Graph Databases

NoSQL Graph Stores

- Represent data in **graphical structures**: Nodes and Edges.
- **Nodes** represent **entities**, **Edges** represent **relationships** between entities.
- Relationships are **directed**, semantics of the direction is up to the application. E.g. "Married" is reflexive, "Owns" is not.
- Each Node/Edge has a set of **Key/Value properties**
- Each Node/Edge has a **label** (type of entity/relationship)
- **Optimized** to process **graph-related queries** and analytics.



- Example Tools
  - Neo4j
  - OrientDB
  - Titan
  - Apache Giraph
  - Microsoft Graph Engine (Trinity)

Real-world Scenarios

- Social Networks
- Network and IT Operations
- Fraud Detection
- Digital Assets Management

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Graph Databases

NoSQL Graph Stores

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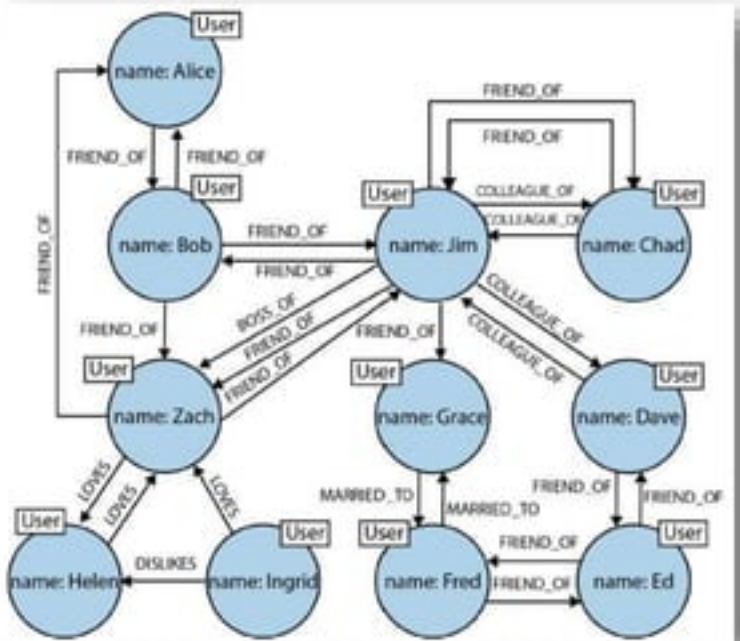


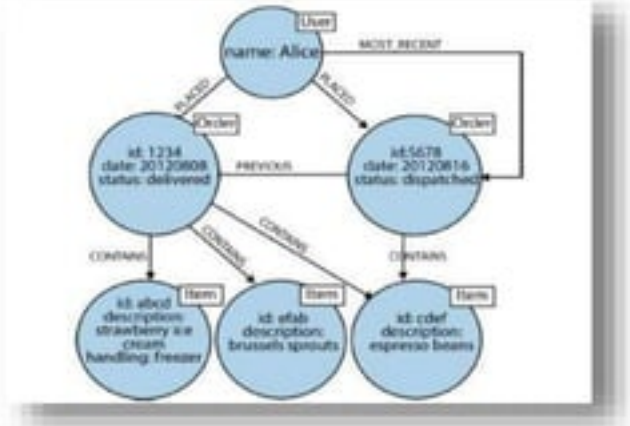


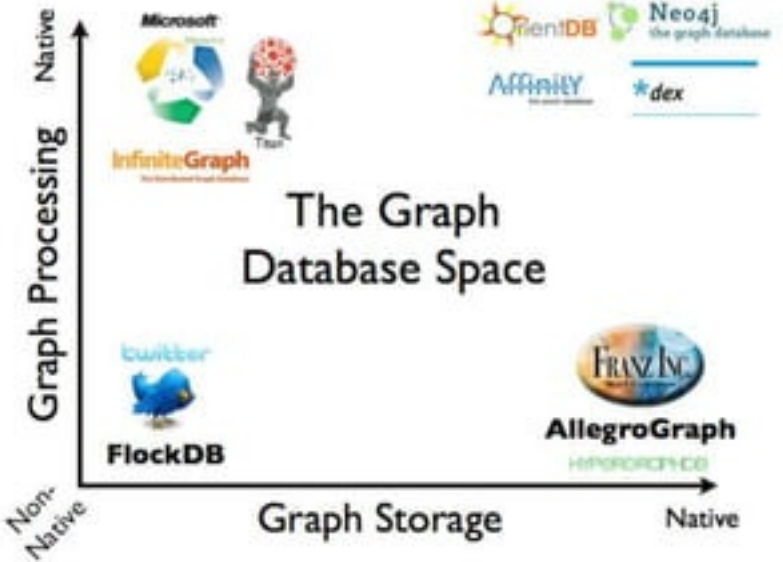
Table 2-1. Finding extended friends in a relational database versus efficient finding in Neo4j

Depth	RDBMS execution time(s)	Neo4j execution time(s)	Records returned
2	0.016	0.01	~2500
3	30.267	0.168	~110,000
4	1543.505	1.359	~600,000
5	Unfinished	2.132	~800,000

Graph Databases

NoSQL Graph Stores

index-free adjacency; connected nodes physically "point" to each other in the database



Any database *behaves* like a graphDB; exposes a graph data model through CRUD operations

Graphs are serialized in any database; Relational, Document, or objectDBs

Storage is designed and optimize to store, process, and query graph data structures

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Neo4j Graph Database

- Most Popular GraphDB (according to db-engines).
- Free Community Edition and Commercial Enterprise Edition.
- Native Graph Processing and Storage.
- Uses Cypher Query Language (CQL).
- Scalability (Redundancy and Load Balancing) with High Availability (HA) package.
- Read capacity of HA cluster increases linearly with the number of servers.
- Can commit 10K of writes per second while maintaining fully ACID transactions.

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Applied Graph Analytics

Neo4j Graph Database

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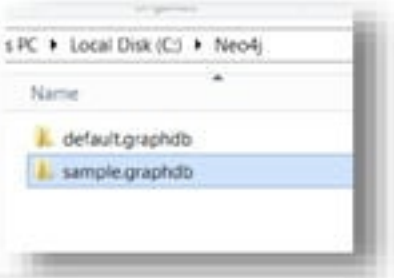
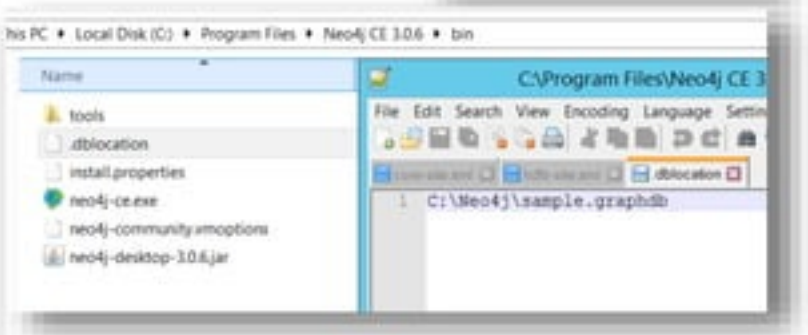
Create a Database

1. Create a folder in your file system (e.g. sample.graphdb)
2. Set the location of the database in .dblocation
3. Lunch neo4j





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Neo4j Graph Database

Create a node

```
create (<Id>:<label>{<Property>:"Value",…})
```

Example

```
create (p1:Person{name:"khalid", age:"31", gender:"male"})
```

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Neo4j Graph Database

Create an edge

create ((<nodeId>)
-[<edgeId>:<label>{<Property>:"Value",…}]->
(<nodeId>))

Example

create ((p1)-[e1:follows{datetime:"2010-10-05"}]->(p2))

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Retrieve nodes/edges

match (<pattern>) return (<objects>)

Example

match (p:perons) return p
Match (p1)-[r]->(p1) return p1,p2, r

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Neo4j Graph Database

Update graph

match (<pattern>) merge (<objects>)

match (<pattern>) set (<object>.property = value)

Example

match (p:perons{name="khalid Salama") merge (p)-[:marriedTo]->(m:perons{name="Fatima Zahra"})

match (p:person) where name = "khalid Salame" set job="IT Manager"

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Neo4j Graph Database

Delete nodes/edges

match (<pattern>) delete (<objects>)

Example

match (n)-[e]-() delete n,e

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Neo4j Graph Database

Import csv Data to Neo4j

Source	Target	Distance
A	B	4
A	C	5
B	D	5
C	B	6

```
LOAD CSV WITH HEADER <filepath>.csv AS line
MERGE (x:city{name:line.Source})
MAEGE (y:city{name:line.Target})
MERGE (x)-[:To{distance=line.Distance}]->(y)
```

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Applied Graph Analytics

Neo4j Graph Database

Basic Queries

```
//Counting the number of nodes
match (n:Label)
return count(n)
//Counting the number of edges
match (n:Label)-[r]->()
return count(r)
//Finding leaf nodes:
match (n:Label)-[r:TO]->(m)
where not ((m)-->())
return m
//Finding root nodes:
match (m)-[r:TO]->(n:Label)
where not (()-->(m))
return m
//Finding triangles:
match (a)-[:TO]->(b)-[:TO]->(c)-[:TO]->(a)
return distinct a, b, c
```

```
//Finding 2nd neighbors of D:
match (a)-[:TO*..2]-(b)
where a.Name='D'
return distinct a, b
//Finding the types of a node:
match (n)
where n.Name = 'Egypt'
return labels(n)
//Finding the label of an edge:
match (n {Name: 'Egypt'})<-[r]-()
return distinct type(r)
//Finding all properties of a node:
match (n:Actor)
return * limit 20
//Finding loops:
match (n)-[r]->(n)
return n, r limit 10
//Finding multigraphs:
match (n)-[r1]->(m), (n)-[r2]-(m)
where r1 <> r2
return n, r1, r2, m limit 10
//Finding the induced subgraph given a set of nodes:
match (n)-[r:TO]-(m)
where n.Name in ['A', 'B', 'C', 'D', 'E'] and m.Name in ['A', 'B', 'C', 'D', 'E']
return n, r, m
```

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Path Analysis

```
//Finding paths between specific nodes:
match p=(a)-[:TO]-(c)
where a.Name='H' and c.Name='P'
return p limit 1
//Finding the length between specific nodes:
match p=(a)-[:TO*]-(c)
where a.Name='H' and c.Name='P'
return length(p) limit 1
//Finding a shortest path between specific nodes:
match p=shortestPath((a)-[:TO*]-(c))
where a.Name='A' and c.Name='P'
return p, length(p) limit 1
//All Shortest Paths with Path Conditions:
match p = allShortestPaths((source)-[r:TO*]->(destination))
where source.Name='A' and destination.Name = 'P' and
length(nodes(p)) > 5
return extract(n in NODES(p)| n.Name) as Path, length(p) as
PathLenght
```

```
//Diameter of the graph:
match (n:Label), (m:Label)
where n <> m
with n, m
match p=shortestPath((n)-[*]->(m))
return n.Name, m.Name, length(p)
order by length(p) desc limit 1
//Extracting and computing with node and properties:
match p=(a)-[:TO*]-(c)
where a.Name='H' and c.Name='P'
return extract(n in nodes(p)|n.Name) as Path, length(p) as pathLength,
reduce(s=0, e in relationships(p)| s + toInt(e.dist)) as pathDist limit 1
//Graph not containing a selected node:
match (n)-[r:TO]->(m)
where n.Name <> 'D' and m.Name <> 'D'
return n, r, m
match (d {Name:'D'})-[:TO]-(b)<-[:TO]-(root)
where not((root)<--())
return (root)
//Graph not containing a selected neighborhood:
match (a {Name: 'F'})-[:TO*..2]-(b)
with collect(distinct b.Name) as MyList
match (n)-[r:TO]->(m)
where not(n.Name in MyList) and not (m.Name in MyList)
return distinct n, r, m
```

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Connectivity Analysis

```
// Find the outdegree of all nodes
match (n:Label)-[r]->()
return n.Name as Node, count(r) as Outdegree
order by Outdegree
union
match (a:Label)-[r]->(leaf)
where not((leaf)-->())
return leaf.Name as Node, 0 as Outdegree
// Find the indegree of all nodes
match (n:Label)<-[r]-()
return n.Name as Node, count(r) as Indegree
order by Indegree
union
match (a:Label)<-[r]-(root)
where not((root)<--())
return root.Name as Node, 0 as Indegree
```

```
//Find the degree of all nodes
match (n:Label)-[r]-()
return n.Name, count(distinct r) as degree
order by degree
// Find degree histogram of the graph
match (n:Label)-[r]-()
with n as nodes, count(distinct r) as degree
return degree, count(nodes) order by degree asc
//Save the degree of the node as a new node property
match (n:Label)-[r]-()
with n, count(distinct r) as degree
set n.deg = degree
return n.Name, n.deg
// Construct the Adjacency Matrix of the graph
match (n:Label), (m:Label)
return n.Name, m.Name,
case
when (n)-->(m) then 1
else 0
end as value
```

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Neo4j Graph Database - Example

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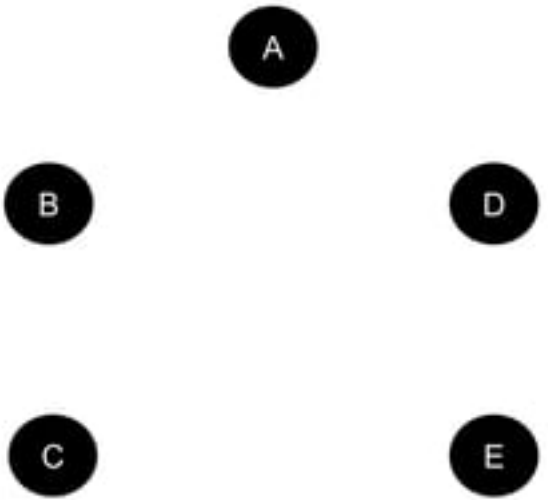

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Neo4j Graph Database - Example

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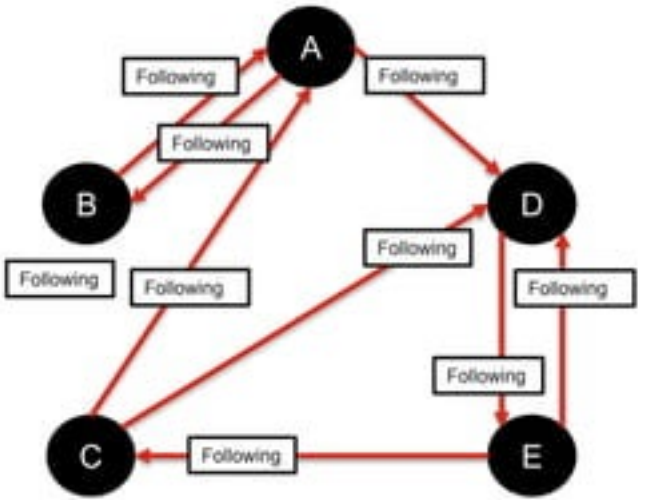

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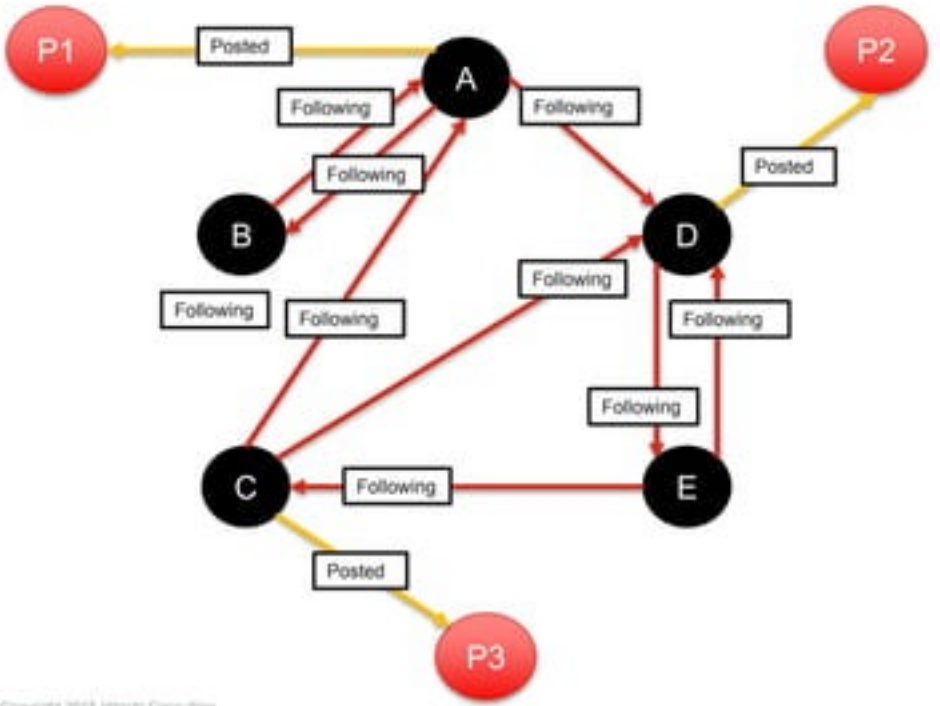

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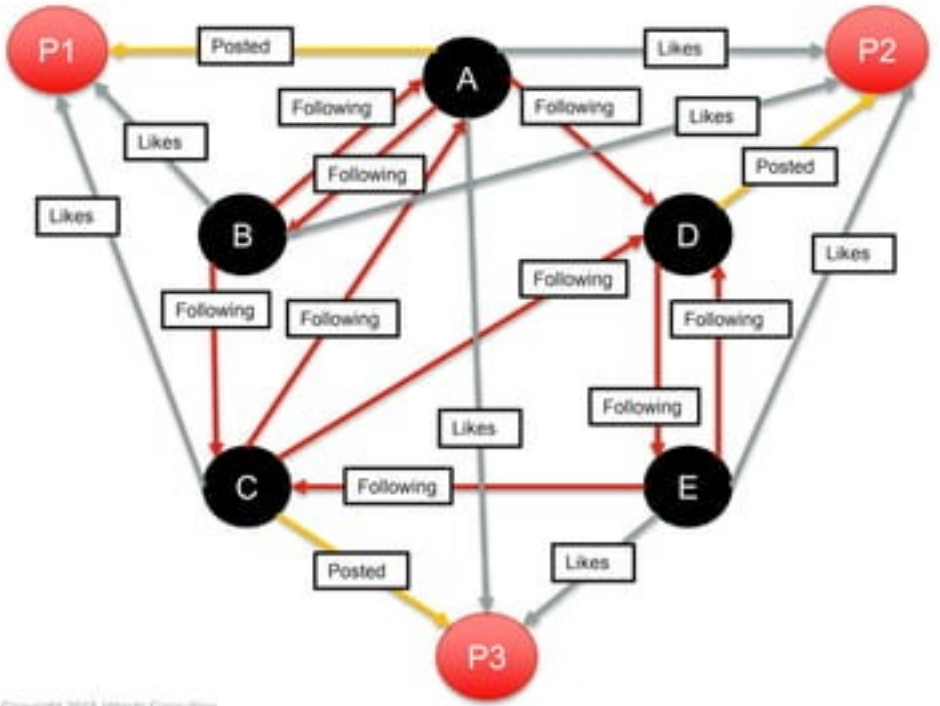

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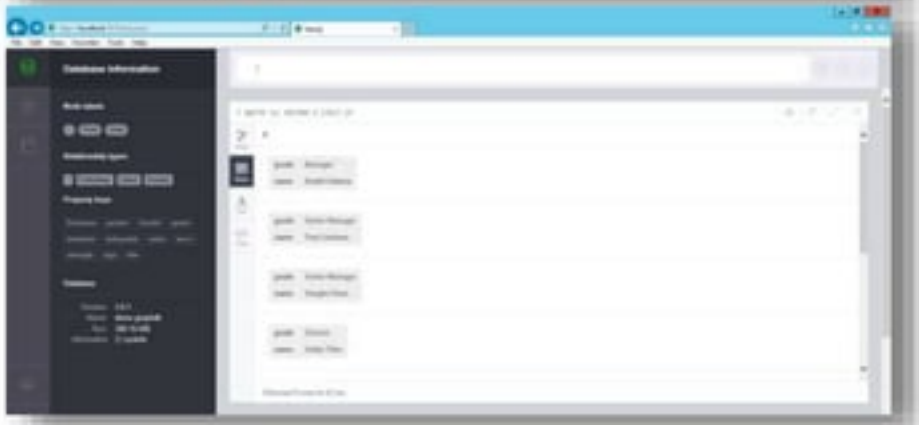
Neo4j Graph Database - Example

```
CREATE

(a:User{name:"Khalid Salama", grade:"Manager"}),
(b:User{name:"Paul Lineham", grade:"Senior Manager"}),
(c:User{name:"Vaughn Rees", grade:"Senior Manager"}),
(d:User{name:"Sutha Thiru", grade:"Director"}),
(e:User{name:"Mark Hill", grade:"VP"}),

(a)-[:Following{since:'2014'}]->(d),
(a)-[:Following{since:'2014'}]->(b),
(b)-[:Following{since:'2010'}]->(a),
(d)-[:Following{since:'2011', strength:"high"}]->(e),
(e)-[:Following{since:'2014'}]->(d),
(e)-[:Following{since:'2015'}]->(c),
(c)-[:Following]->(d),
(c)-[:Following{since:'2013', strength:"low"}]->(a),
(b)-[:Following]->(c),

(p1:Post{title:"post 1", lastupdate:"01/01/2016", tags:['sports','life style']}),
(p2:Post{title:"post 2", lastupdate:"03/05/2015"}),
(p3:Post{title:"post 3", lastupdate:"121/7/2015", tags:['economics','politcs']}),

(a)-[:Posted]->(p1),
(d)-[:Posted]->(p2),
(c)-[:Posted]->(p3),

(b)-[:Liked]->(p1),
(c)-[:Liked]->(p1),
(a)-[:Liked]->(p2),
(b)-[:Liked]->(p2),
(e)-[:Liked]->(p2),
(a)-[:Liked]->(p3),
(e)-[:Liked]->(p3)
```



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Neo4j Graph Database - Example

```
//fetch one node
MATCH (u:User{name:"Khalid Salama"}) RETURN u

// fetch an attribute of a node
MATCH (u:User{name:"Khalid Salama"}) RETURN u.grade

// fetch nodes by conditions
MATCH (u:User{grade:"Senior Manager"}) RETURN u
--
MATCH (u:User)
WHERE u.grade = 'Senior Manager'
RETURN u
--
MATCH (u:User)
WHERE u.name =~ "Sutha.*" // START WITH, END WITH, CONTAIN, IN [,].
RETURN u
--
MATCH ()-[r:Posted]->(p:Post)
WHERE 'sports' IN p.tags
RETURN p

// Whom khalid is following?
MATCH (x:User{name:"Khalid Salama"})-[r:Following]->(y:User)
RETURN x,r,y

// Who is Following Khalid
MATCH (x:User{name:"Khalid Salama"})<-[r:Following]-(y:User)
RETURN x,r,y

// Update
MERGE (u:User { name:"Khalid Salama" })
SET u.practice = "Data Insights & Analytics"
RETURN u
```



```
// Get Count of Posts
MATCH (p:Post) RETURN COUNT(p)
-- Get User Count By Grade
MATCH (u:User) RETURN u.grade, COUNT(u)
-- Get User and Followers
MATCH (u:User)<-[:Following]-(f:User)
RETURN u.name AS User,COLLECT(f.name) AS follows,COUNT(f) AS Total

// Constraint
CREATE CONSTRAINT ON (u:User) ASSERT u.name IS UNIQUE
-- Index
CREATE INDEX ON :User(grade)

// Get users following each other
MATCH (u1:User)-[:Following]->(u2:User)-[:Following]->(u1)
RETURN u1.name,u2.name

// Get Users likes a post posted by a follower
MATCH (u:User)-[:Liked]->(p:Post)<-[:Posted]-(u2:User)-[:Following]->(u)
RETURN u,p,u2

// Get Following of Following
MATCH (u:User)-[:Following]->()-[:Following]->(u2:User)
Return u.name,COLLECT(DISTINCT u2.name)

// Get User with max 3 steps from Paul
MATCH (u:User)-[:Following*..3]->(us:User{name:"Paul Lineham"})
Return u

// Shortest path

MATCH
        (u1:User{name:"Mark Hill"}),
        (u2:User{name:"Paul Lineham"}),
        p=SHORTESTPATH((u1)-[:Following*..10]->(u2))
RETURN p

-- Get nodes having a property
MATCH(p)
WHERE EXSITS(p.tags)
```

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http://neo4j.com/docs/developer-manual/current/#cypher-query-lang

Useful Resources

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- **Coursera** – Graph Analytics for Big Data
https://www.coursera.org/learn/big-data-graph-analytics/home/welcome

- **Coursera** – Data Manipulation at Scale (Lessons 21-24)
https://www.coursera.org/learn/data-manipulation/home/week/4

- **Neo4j** – Getting Started Tutorials
https://neo4j.com/developer/get-started

- **Apache Spark** – GraphX Documentation
http://spark.apache.org/docs/latest/graphx-programming-guide.html

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My Background

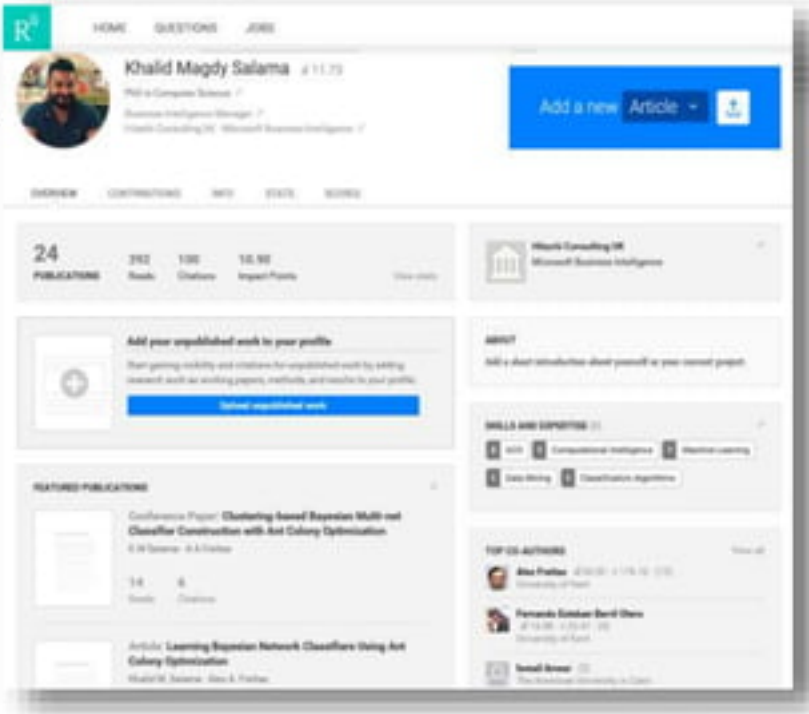
Applying Computational Intelligence in Data Mining

- Honorary Research Fellow, School of Computing , University of Kent.
- Ph.D. Computer Science, University of Kent, Canterbury, UK.
- M.Sc. Computer Science , The American University in Cairo, Egypt.

- 25+ published journal and conference papers, focusing on:
  - *classification rules induction,*
  - *decision trees construction,*
  - *Bayesian classification modelling,*
  - *data reduction,*
  - *instance-based learning,*
  - *evolving neural networks*, and
  - *data clustering*

- **Journals**: *Swarm Intelligence, Swarm & Evolutionary Computation, , Applied Soft Computing, and Memetic Computing.*

- **Conferences**: *ANTS, IEEE CEC, IEEE SIS, EvoBio, ECTA, IEEE WCCI and INNS-BigData.*



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Thank you!

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